Wire Bond Failure Analysis During the Molding Process

1. Problem Statement

In this case, I will consider a scenario where wire bond failures increase significantly during backend assembly. The failure rate will increase from around 1% to approximately 4.5%, and the failures will be observed during final electrical testing as **open circuit failures or Intermittent failure.**

My objective is to identify

a. Where the failure is introduced in the process flow
b. Determine the most probable root cause using structured engineering thinking.

2. Creating and Analyzing the Dataset (Data-Driven Thinking)

Creating and Analyzing the Dataset

Before moving to physical failure analysis, I first approached the problem from a **data perspective**, as yield issues in a real OSAT manufacturing environment are always identified and tracked through production data.

Since actual factory data is not available, I created a **simulated production-like dataset** that mimics realistic lot-wise manufacturing behavior.

How the Dataset Was Created

In semiconductor manufacturing, production is organized into **lots (batches)**, each of which consists of multiple units processed under similar process conditions.

Based on this, I constructed a dataset with the following characteristics:

- Constant number of units per lot (**1000 units**)
- Stable baseline failure rate (~1%) representing normal process conditions
- Sudden increase in failure rate (~4–5%), indicating a potential process issue
- Subsequent recovery to normal levels, simulating corrective action

Lot No	Total Units	Failures Units	Failure %
L1	1000	10	1
L2	1000	12	1.2
L3	1000	45	4.5
L4	1000	42	4.2
L5	1000	48	4.8
L6	1000	15	1.5
L7	1000	11	1.1
L8	1000	13	1.3

Table 2.1: Dataset for the failure

Note:
A **lot** is defined as a batch of semiconductor units processed together under the same manufacturing conditions, and it serves as the fundamental unit for yield tracking and analysis.

Data Analysis and Observations

From the dataset, the following observations can be made:

- Lots L1 and L2 show stable failure rates (~1%) → indicating normal process conditions
- Lots L3 to L5 show a significant increase in failure rate (~4–5%)
- From Lot L6 onwards, a reduction in failure rate is observed, indicating potential improvement **due to process optimization or corrective actions.**

To better visualize this behavior, a **Failure Rate (%) vs Lot** graph was plotted.

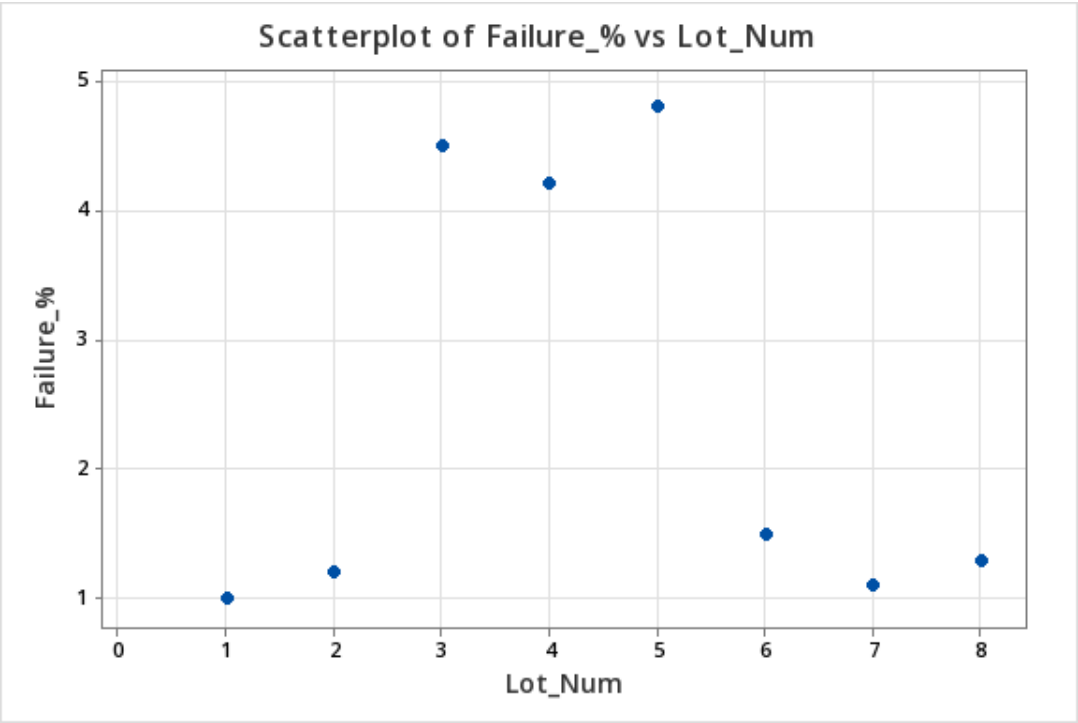

Figure: ***Failure Rate (%) vs Lot***

Trend Interpretation

The graph clearly shows:

- A stable region (L1–L2)
- A sharp increase in failure rate (L3–L5)
- A recovery phase (L6–L8)

This pattern is characteristic of a **process excursion**, in which a temporary deviation in process conditions leads to an increase in defects, followed by correction.

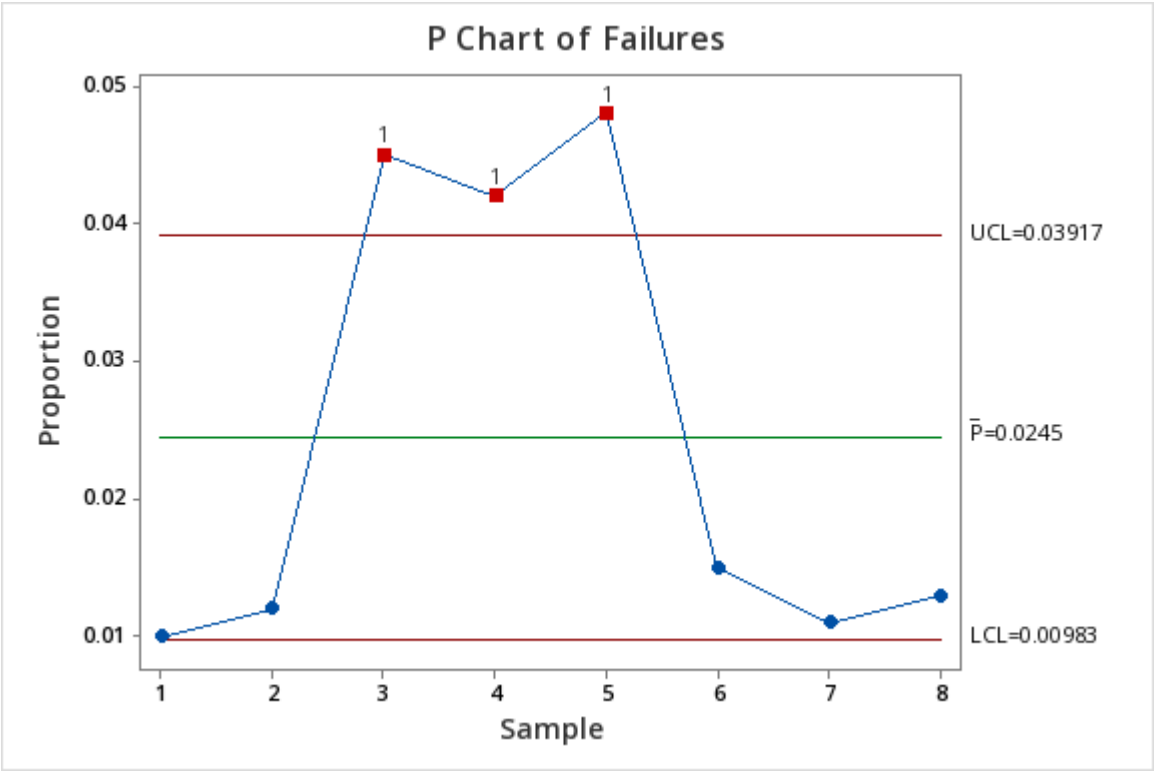

Figure: P-Chart of Failure Proportion

P-Chart Interpretation:

- Lots L3 to L5 exceed the Upper Control Limit (UCL)
- This indicates that the process was out of statistical control during this period
- The remaining lots fall within control limits, confirming normal operation

This confirms that:

The increase in failure rate is due to a specific assignable cause, not random variation.

Engineering Interpretation

At this stage, the key question becomes: What does a sudden increase in failure rate indicate?

In semiconductor manufacturing:

- Gradual variation → normal process fluctuation
- Sudden jump → process excursion or parameter change

The observed behavior strongly suggests that:

- A process parameter changed around Lot L3
- The issue persisted for a few lots
- Corrective action restored the process after Lot L5

Key Insight:

The failure is not random; it is directly linked to a temporary change in process conditions starting from Lot L3.

Next Logical Question:

What changed around Lot L3?

Possible causes include:

- Machine parameter variation
- Change in molding pressure or flow conditions
- Material batch variation (mold compound)
- Operator or setup variation

Importance of a Data-Driven Approach

This dataset enables:

- Identification of the exact point where the issue started

- Isolation of the affected production window (L3–L5)
- Focused investigation on the relevant process stage
- Strong support for root cause analysis

Summary:

Created a simulated lot-wise dataset representing real production behavior

- Maintained:
    - Stable baseline (~1%)
    - Sudden failure increase (~4–5%)
    - Recovery phase (process correction)
- Identified:
    - Process shift starting from Lot L3
- Validated:
    - Process is out of control using the P-chart
- Interpreted:
    - Sudden variation → process excursion (not random)
- Used data to:
    - Narrow down the investigation scope
    - Guide root cause analysis

After identifying the root cause and implementing corrective actions, the failure rate shows a decreasing trend across subsequent lots.(from L6)

This behavior is consistent with real manufacturing environments, where process adjustments lead to gradual yield recovery.

3. How I Will Approach the Problem

When I first see this issue on the production floor, I won't jump to conclusions immediately. Instead, I will try to understand the problem step by step, just as a process engineer would handle a real yield excursion.

When analyzing the failure, I would also look at the test data from the Automatic Test Equipment (ATE). The ATE provides detailed information about the type of electrical failure observed.

For an open failure, the ATE typically shows very high resistance or no current flow in the affected path. The test logs may indicate conditions like "open connection" or abnormal voltage levels due to a lack of continuity.

For a short failure, the ATE would show unintended current flow between nodes, often reflected as low resistance or excessive current. The test results may indicate "short circuit" or leakage beyond acceptable limits.

In some cases, **intermittent failures** may also be observed, where the connection is unstable and passes or fails under different conditions. These are usually more challenging and may require further stress testing or detailed analysis.

By carefully analyzing ATE logs and failure patterns, I can accurately classify the failure type, which is a critical first step before moving into physical and process-level investigation.

I would begin by looking at the failure reports from the final test. Since the failure rate has suddenly increased, my first thought would be to understand what kind of failure it is. If it is an open circuit, it clearly indicates that somewhere the electrical connection is getting broken.

Once I know the failure type, the next thing I would try to figure out is at which stage this failure is getting introduced. I would mentally walk through the entire assembly flow, from die attach, wire bonding, molding, to final test, and try to correlate where this kind of failure can originate.

If I observe that the units are passing earlier stages and only failing after molding, then I can narrow down my focus to the molding process. This helps me avoid wasting time investigating unrelated processes.

After identifying the stage, I would start thinking about what physical mechanisms can actually cause this type of failure. If I assume it is an open failure, since it is an open failure and we are dealing with bond wires, I would consider possibilities like mechanical stress, deformation, or breakage of the wire.

At this point, I would connect this understanding with actual process conditions. I know that during molding, high pressure and compound flow are involved, which can exert force on delicate bond wires. So I would start evaluating whether molding parameters such as pressure, flow behavior, or wire loop height could be contributing to the issue.

So overall, my approach is not to assume the root cause directly, but to move step by step, from failure type to failure stage to physical mechanism, and finally to process correlation. This structured way helps in identifying the root cause accurately and efficiently.

Summary of Approach:

- Analyze failure using ATE test data and logs
- Identify the failure type (open/short/intermittent)
- Determine the stage where failure is introduced
- Correlate failure with process flow
- Evaluate possible physical mechanisms causing the failure
- Link failure mechanism with process conditions (e.g., molding parameters)
- Narrow down to the most probable root cause using elimination logic

3. Understanding the Failure Type

Once the failure is reported from production, the first thing I would do is check the ATE test logs to understand the nature of the failure.

From the ATE data, if I observe no current flow or very high resistance, I can classify the failure as an open circuit. In some cases, the test log may directly indicate an “open” condition or show abnormal voltage behavior due to loss of continuity.

At this point, I clearly understand that:

The failure is an open, which means there is a break in the electrical path somewhere inside the package.

Now I start asking myself:
Where can an open occur in a package?

Instead of guessing, I break it down systematically.

Possible locations for open failure:

- Die pad connection (die-to-substrate attach issue)
- Bond wire (wire break or weak bond)
- Lead frame or substrate routing
- Solder joint (ball attach issue)

Narrowing Down Based on Package Type

Since this is a wire-bonded package, I immediately give more attention to the bond wire region because:

- Bond wires are mechanically fragile
- They are exposed to process stresses (especially during molding)
- They are a common failure point in assembly

Engineering Insight:

At this stage, I don’t conclude anything yet.

But I built a working hypothesis:

There is a high probability that the open is related to bond wire integrity, but I need to confirm at which stage this damage is happening.

Summary:

- Analyze ATE logs to classify failure type
- Identify electrical signature:
    a. Open → High resistance / no current flow
- Ask: Where can an open occur in a package?
- List all possible locations:
    a. Die pad
    b. Bond wire
    c. Substrate
    d. Solder joint
- Focus on the bond wire (the most sensitive component)
- Form an initial hypothesis without jumping to a conclusion

4. Identifying the Failure Stage (Critical Step in Analysis)

After understanding that the failure is an open circuit, the next step is to determine at which stage this failure is being introduced in the assembly flow.

Instead of randomly checking all processes, I would approach this logically by correlating the failure with different stages of the manufacturing flow.

First, I visualize the process flow:

Wafer → Die Attach → Wire Bond → Molding → Ball Attach → Final Test

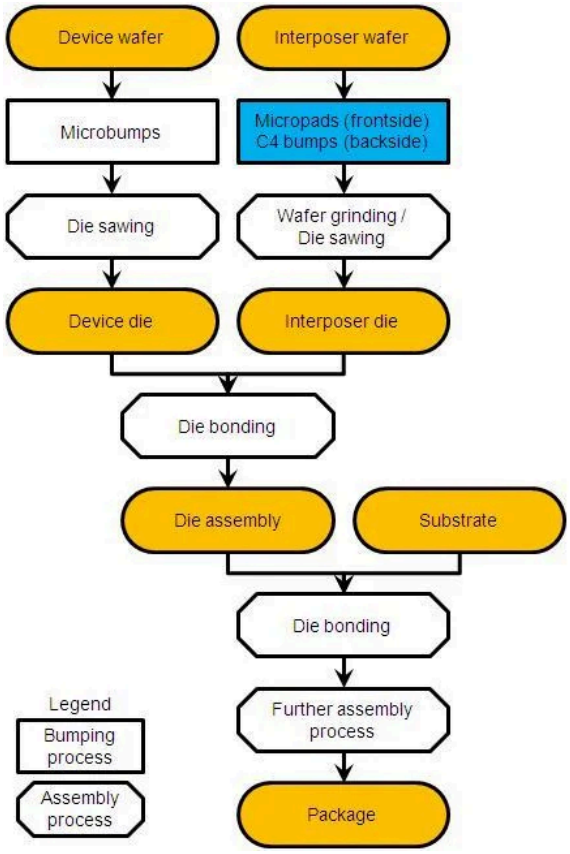

Figure 4.1: Typical semiconductor backend assembly process flow

Now I ask myself: At which stage could this open failure realistically occur?

Step 1: Check if failure existed before molding

If the issue is related to:

- Weak bonding
- Capillary defect
- Improper bonding parameters

Then the failure should appear:

- During in-line inspection
- Or before molding
- Or in early electrical checks

But observation:

- Units are passing earlier stages
- Failure appears only at the final test (after molding)

So I eliminate:

- Bonding defect (primary cause)
- Capillary/tooling issue (early failure expected)

Step 2: Focus on processes after bonding

Now I shift my focus to:

- Molding
- Ball attach
- Handling / mechanical stress

Evaluate each:

a. Ball Attach:
    - Mostly affects solder joints

- Less likely to break the bond wire
  Less probable

b. Handling / Transport:

- Can cause damage, but usually random
- Not a consistent lot-wise pattern
  Less likely

c. Molding Process:

- High pressure
- High temperature
- Mold compound flow
  Can introduce mechanical stress on bond wires
  Highly probable

Key Insight

The failure appears after molding, and molding is the first process where:

- External mechanical force acts on bond wires
- Flow dynamics can physically disturb wires

Conclusion of This Step:

The failure is most likely introduced during the molding process, based on stage correlation and elimination of upstream processes.

Summary (Point-wise)

1. Map the full assembly process flow

2. Ask: At which stage can this failure occur?

3. Check if failure exists before molding

4. Eliminate:

   4.1. Bonding issues (would fail earlier)

4.2. Tooling defects (early detection expected)

5. Evaluate downstream processes:

5.1. Ball attach → low probability

5.2. Handling → random, not consistent

5.3. Molding → high mechanical stress

6. Correlate failure timing with process stage

7. Conclude: Failure introduced during molding

5. Physical Failure Mechanism (What Actually Happened to the Wire)

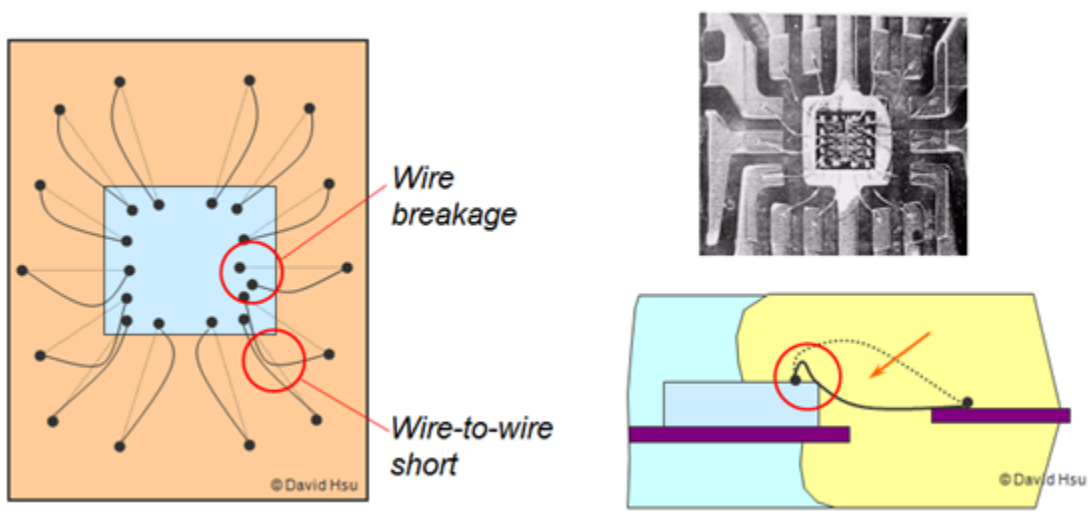

Figure 5.1: Wire sweep and bond wire deformation during molding

After identifying that the failure is introduced during the molding process, the next step is to understand:

What exactly happens physically to cause the wire to break?

First, I think about molding conditions

During molding, the package is exposed to:

- High pressure
- High temperature
- Flow of epoxy molding compound

This is the first stage where external forces act on bond wires

Now I ask:
How can molding physically damage a bond wire?

Key Observation

Bond wires are:

- Very thin
- Mechanically delicate
- Suspended in air (loop structure)

This makes them vulnerable to external forces

🧠 Step-by-Step What Happens

1. Mold compound enters the cavity
2. Compound flows at high velocity
3. Flow hits the bond wires
4. The wire gets pushed sideways (wire sweep)
5. Stress builds, especially at:
    - Neck region (near the bond point)
6. During cooling:

    - Additional stress due to thermal mismatch

7. The wire weakens and eventually breaks

Critical Factors That Make It Worse

I would evaluate:

- High molding pressure → increases force on wire
- High loop height → more exposed to flow
- Low wire strength → easier to break
- Flow direction/velocity → uneven stress

Engineering Insight

This is not always an instant failure.

👉 Sometimes:

- The wire gets weakened during molding
- Breaks happen later during cooling or handling

What I Would Expect to See

If I inspect physically:

- X-ray → broken or displaced wire
- SEM → necking or fracture point
- Optical → wire deformation

6. Final Understanding

The bond wire experiences mechanical displacement due to mold compound flow (wire sweep), leading to stress concentration and eventual fracture.

Summary (Point-wise)

- Molding introduces:
    a. High pressure
    b. High temperature
    c. Compound flow
- Bond wires are:
    - Thin and mechanically weak
- Mold flow causes:
    - Wire sweep (lateral displacement)
- Stress builds at:
    - Neck region
- Additional stress during:
    - Cooling (CTE mismatch)
- Final result:
    - Wire break → Open failure

7. Root Cause Confirmation & Process Improvement

Based on the systematic analysis carried out in the previous sections, including failure stage correlation, physical mechanism evaluation, process parameter comparison, and validation through controlled adjustments, the root cause of the failure can be clearly identified.

Root Cause

The primary root cause of the observed failure is wire sweep during the molding process, caused by:

- Excessive molding pressure
- Suboptimal wire loop height

These conditions result in significant mechanical stress on the bond wires, particularly near the neck region, leading to deformation and eventual wire breakage. This manifests as an open circuit failure during final electrical testing.

Corrective Actions

To address the identified issue, the following corrective actions are proposed:

- Optimization of molding pressure to reduce the mechanical force exerted by mold compound flow
- Reduction of wire loop height to improve structural stability and minimize exposure to flow-induced forces
- Improved monitoring and control of molding process parameters to maintain process consistency

Result and Validation

After implementing the above corrective actions (simulated scenario), a clear improvement in yield performance is observed:

- Failure rate reduced from approximately 4–5% (Lots L3–L5)
- Restored to approximately 1–1.5% from Lot L6 onwards

This recovery trend indicates that the process has returned to a stable operating condition.

Final Insight

The observed behavior represents a typical yield excursion followed by recovery, which is commonly seen in semiconductor manufacturing when process deviations are identified and corrected.

This confirms that:

The failure was caused by molding-induced wire sweep, and the implemented process optimizations effectively mitigated the issue.

7. Root Cause Conclusion

Based on:

- Failure stage correlation
- Physical mechanism analysis
- Process parameter comparison
- Validation through controlled adjustments

Conclusion

The root cause of the observed failure is identified as **wire sweep during the molding process**, primarily caused by excessive molding pressure and suboptimal wire loop height. This leads to mechanical stress on the bond wires, resulting in wire break and open circuit failure.

Corrective Action and Result

After optimizing molding pressure and reducing loop height (simulated scenario), the failure rate decreased from approximately **4–5% (Lots L3–L5)** to around **1–1.5% from Lot L6 onwards**, indicating successful process stabilization.

References:

The analysis and conclusions in this case study are based on standard semiconductor packaging principles and widely accepted failure mechanisms. The following resources were referred to for conceptual understanding:

- Fundamentals of Microsystems Packaging — *Rao R. Tummala*
  For semiconductor packaging fundamentals, molding process, and reliability concepts
- NPTEL Electronics Systems Packaging — *Prof. K. N. Bhat*
  For backend assembly flow and packaging processes

- Microelectronics Packaging Handbook — *Rao R. Tummala*
  For wire bonding, molding behavior, and packaging technologies
- Microelectronics Failure Analysis Desk Reference — *Ed. EDFAS (ASM International)*
  For failure mechanisms related to open circuits and wire bond breakage
- Illustrative images and diagrams are adapted from standard semiconductor packaging resources and publicly available educational materials